

Wyatt Brege

Expertise

- **recent open-source projects**
 - **oshea** – a Node.JS Markdown to PDF converter and extensible doctype plugin system
 - **aoife** – a simple multi-media poster sharing matrix made with React
 - **thunder-muscle** – a data exporter and python API for Thunderbird. Basis for a project to data-mine decades of email.
 - **ilma** – a pure-shell project-mirror supporting context generation, repo stats, project pruning, advanced configurable archival, and natively supported by `tar`, `rsync`, `gpg` in one convenient wrapper
 - **the-flavor-network** – an interactive creativity tool for chefs and bartenders to visualize recipes through flavor pairings
- simulated complex physical systems with terabytes of data on supercomputing clusters: **SpEC**, stampede, xsede, calcul cacr.caltech
- contributed to large, collaborative driven codebases with: **node.js**, **python**, **c++**, **git**, **javascript**, **pandas**, **networkx**, **fortran**, **tensorflow**, **perl**, **bash**, **php**, **sql**, **numpy**, **matplotlib**, **d3.js**, **hugo**, **gnuplot**, **latex**, **paraview**, **sage**, **jupyter**
- advanced physics and mathematics specialties: mechanics, general relativity, **fluid dynamics**, **finite volume**, numerical methods
- developed tests, validation and visualization tools of big data

Education

PhD, *Washington State University*, Physics, 2017

BSc, *Grand Valley State University*, Mathematics, 2010

BSc, *Grand Valley State University*, Physics, 2010

ASc, *Grand Rapids Community College*, 2007

Professional

Data Science Fellowship, *The Data Incubator*, San Francisco, CA, 2023

Restaurant Consulting, CA, MI, 2020-2022

Executive Chef, *Slanted Tree*, Fairfield, CA, 2019-2020

Executive Chef, *Dawson's*, Dixon, CA, 2017-2019

Research Assistant, *Washington State Univ*, Pullman, WA, 2013-2017

Teaching Assistant, *Washington State Univ*, Pullman, WA, 2010-2017

Sous Chef, *One Trick Pony*, Grand Rapids, MI, 2005-2007

Publications

Chakravarti, K., Gupta, A., Bose, S., Duez, M.D., Caro, J., **Brege, W.**, Foucart, F., Ghosh, S., Kyutoku, K., Lackey, B.D. and Shibata, M. *Systematic effects from black hole-neutron star waveform model uncertainties on the neutron star equation of state*, January 31, 2019 Physical Review D. 99(2), 024049 *preprint*: arXiv:1809.04349

Brege, W., Duez, M.D., Deaton, M.B., Foucart, F., Caro, J., Hemberger, D.A., Kidder, L.E., O'Connor, E., Pfeiffer, H.P., Scheel, M.A. *Black hole-neutron star mergers using a survey of finite-temperature equations of state*, September 12, 2018 Physical Review D, 98(6), 063009 *preprint*: arXiv:1804.09823

Foucart, F., Desai, D., **Brege, W.**, Duez, M.D. 4, Kasen, D., Hemberger, D.A., Kidder, L.E., Pfeiffer, H.P., Scheel, M.A., *Dynamical ejecta from precessing neutron star-black hole mergers with a hot, nuclear-theory based equation of state*, November 3, 2016 Classical and Quantum Gravity 34(4) 044002 *preprint*: arXiv:1611.01159

Contributed Talks

Brege, W. *The Flavor Network*, Specially selected for The Data Incubator's Capstone Showcase, November 9, 2023

Brege, W., Foucart, F., Duez, M.D., *Equation of state survey of black hole-neutron star mergers*, APS April Meeting, April 16, 2016, BAPS.2016.APR.C14.3

Brege, W., Duez, M.D., *A high order accurate finite difference scheme with boundary closures for astrophysical simulations*, Northwest APS Meeting, May 16, 2015, BAPS.2015.NWS.E6.6

Brege, W., Duez, M.D., *A stable high-order multipatch method for black hole accretion simulations*, APS April Meeting, April 12, 2015, BAPS.2015.APR.K13.9

Brege, W., Bolen, B., *Polymer quantum mechanics and an approach to minimal length* (Poster), 19th International Conference on General Relativity and Gravitation (GR19), Mexico City, Mexico, July 16, 2010

Brege, W., Brennan, J., *Symmetry Analysis and the Lane-Emden Equation*, 2009 Undergraduate Symposium at Argonne National Labs, Argonne, IL, November 13, 2009

Brege, W., Brennan, J., *Symmetry Analysis of the Lane-Emden Equation* MathFest 2009, Portland, OR, August 07, 2009

Awards

- NASA Space Grant, 2015, 2016 and 2017
- Graduate Assistance in Areas of National Need (GAANN) fellowship, 2010-2011 and 2011-2012
 - taught a multiweek Hamiltonian mechanics lecture series to junior-level physics majors
 - taught engine cycles and thermodynamics to introductory physics students
- Science, Mathematics, And Research for Transformation (SMART) scholarship, 2009-2010
- Outstanding Student Achievement Award, GVSU Mathematics Department, 2009

Research Experience

May 2013 - Sep 2018 **Black hole-neutron star mergers and accretion disk simulations** *Washington State University, SXS collaboration*

- evolved black hole-neutron star systems in SpEC with adaptive mesh refinement and nuclear-theory based equations of state
- implemented a high-order accurate finite difference scheme with boundary closures to solve the fluid equations of an accretion disk on a multipatch grid structure
- completion of dissertation

May 2010 - Aug 2010 **Generalized uncertainty principle and minimal length** *Grand Valley State University*

- determined new connections between polymer quantum mechanics and minimal length
- first time presenting internationally

Aug 2009 - May 2010 **Motion in two-center gravitational systems** *Grand Valley State University*

- implemented Gragg extrapolation to evolve Hamilton's equations for a three-body gravitational system
- demonstrated the chaotic behavior of particle trajectories around generic rotating binary systems
- completion of physics senior thesis

Aug 2009 - Dec 2009 **Quasicrystals, tilings and diffraction patterns** *Grand Valley State University*

- mathematical study of diffraction in aperiodic tiles and quasicrystals
- completion of mathematics senior thesis

May 2009 - Aug 2009 **Symmetry analysis of differential equations** *University of Central Florida*

- determined the underlining Lie symmetries of the Lane-Emden equation
- participated in the UCF combined math and physics Research Experience for Undergraduates

Memberships

- SXS Collaboration, c. 2013
- American Physical Society, c. 2009
- Mathematical Association of America, c. 2009
- Omicron Delta Kappa, c. 2010
- Society of Physics Students (SPS), c. 2009
- Pi Mu Epsilon (Iota chapter), c. 2009